

ACHIEVING CUSTOMER LOYALTY IN RYANAIR: A CRM DRIVEN APPROACH

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PART – 1 SPECIFICATION

CHAPTER 1

Project Summary

The air transport sector plays a critical role in facilitating long and fast mobility for people across the world (Botlíková, Botlík and Stuchlíková, 2021, p. 01005). It is also a vital contributor to the European economy, employing between 1.4-2 million people directly and supporting around 5 million other jobs, both directly and indirectly. According to the European Commission, aviation represents more than €110 billion of Europe's GDP, highlighting the significant impact of the industry.

Despite the significant role played by several aviation companies, in 2019, the biggest five airlines - Lufthansa Group, Ryanair, IAG, easyJet, and Air France-KLM - own 54% of the total seat capacity of the market, emphasizing the competitive nature of the industry. Ryanair, being one of the major players in the industry, aims to reduce costs while increasing the efficiency of invested resources (O'Connell and Williams, 2005, pp.259-272; Malighetti, Paleari and Redondi, 2009, pp.195-203).

Established in 1984, Ryanair has managed to sustainably grow, overcoming various global challenges such as wars, financial crises, and pandemics. The company's competitive advantage lies in its ability to provide affordable fares by minimizing expenses. Ryanair's unique approach has led to it being recognized as Europe's largest low-cost airline, running a fleet of over 450 aircraft. The company is focused on rebuilding Europe's aviation and tourist industries by expanding its passenger network to 225 million by 2026.

This project aims to deliver a consultancy service from Wingspan Consulting for Ryanair, that provides clear insights into its market position and offers a customer relationship management system tailored to the current moment. Through this project, Ryanair will gain a better understanding of its place in the market, which will help it make informed decisions for future growth. Additionally, the customer relationship management system will assist Ryanair in enhancing its customer experience, improve customer retention, and ultimately increase revenue.

This document is organized in two parts. The first part is the specification of the project, and it comprises a background of Ryanair and the European aviation market, the management methodologies applied to the client business, and the system and database designs proposed. The second part is the implementation report, consisting of the technologies implemented, demonstration and explanation of the customer relationship management and dashboards resulting from the application of these technologies, and a final overview of this project's management.

CHAPTER 2

Background and Scope

2.1 Business Model

Ryanair focus on providing low fares for not so demanding customers. It is able to do so by reducing operational expenses, such as offering free food and/or drinks, in-flight entertainment options, comfortable seats, closest airport gates, and other non-essential services (Diaconu, 2012, pp.342-346). The efficiency is another key ingredient for the company financial health, prioritizing the flights schedule and minimizing the ground time of its aircrafts. Altogether, this model has showed that it is solid, and, through the decades of existence, Ryanair built an expertise in operating with low-cost, prevailing in moments in which many airlines bankrupted.

Although, as the costs are the priority, other business concerns fall behind. The customer's feedbacks are, in general, worse than from other companies in the sector. The customer and employee interaction, for example, are discouraged by the company's process. This damaged Ryanair's reputation among passengers. Some of the most common drawbacks for customers include:

- **Additional Fees:**

Ryanair charges a range of additional fees for services that other airlines often include in their ticket prices. This includes fees for checked baggage, priority boarding, seat selection, and even printing boarding passes. These fees can add up quickly and can be a source of frustration for customers who may feel like they are being nickel-and-dimed.

- **Limited Destinations:**

Ryanair primarily operates in Europe, and their route network is more limited than some other airlines. This means that customers may not be able to fly to the exact destination they want or may have to make a connection to reach their destination.

- **Tight Seating Arrangements:**

Ryanair is known for its tight seating arrangements, with little legroom and narrow seats. This can be uncomfortable for some customers, especially on longer flights.

- **Limited Customer Service:**

Ryanair's customer service is often criticized for being slow to respond and unhelpful. This can be frustrating for customers who need assistance with flight changes, cancellations, or other issues.

- **No Frills:**

Ryanair is a no-frills airline, which means that they do not offer many of the amenities that other airlines do. There is no in-flight entertainment, no free snacks or drinks, and no blankets or pillows provided.

- **Strict Baggage Policy:**

Ryanair has a strict baggage policy, which limits customers to one small carry-on bag that must fit under the seat in front of them. This can be challenging for customers who need to travel with more luggage or larger items. Exceeding the baggage allowance can result in additional fees.

So far, the damage from these drawbacks has not been great enough to stop business from grow. From 2011 to 2020, revenue from the budget airline increased by 134%, as can be seen in Fig. 2.1. The abrupt drop in 2021 was due to the coronavirus pandemic.

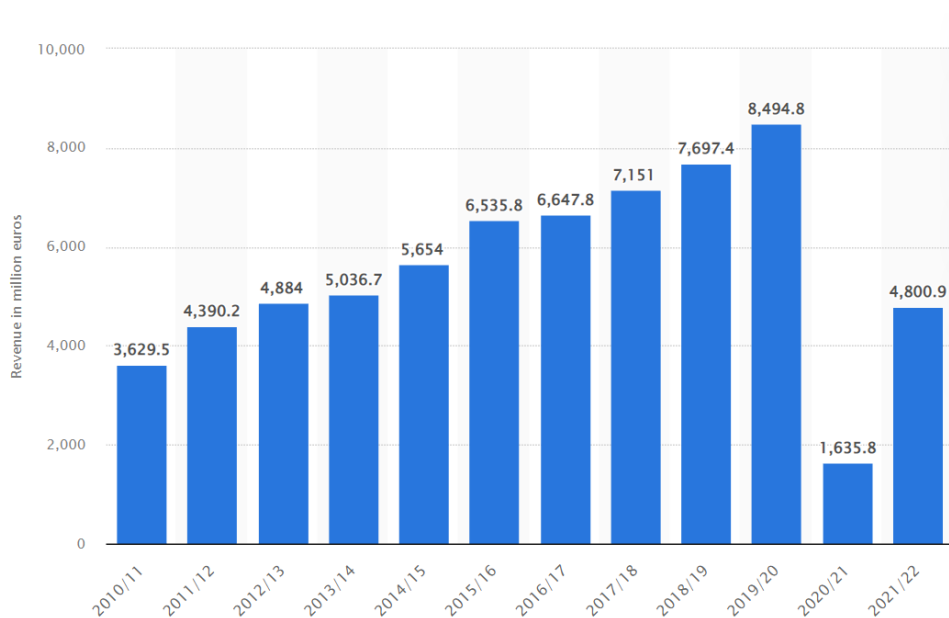


Fig. 2.1 Ryanair's revenue from 2010 to 2022.

Since Ryanair has built its operations around minimizing expenses, it has positioned it as the airline best equipped to navigate the ongoing repercussions of the pandemic. The impacts of the pandemic on Ryanair's revenue are unquestionable, but low-cost approach sets Ryanair apart from its competitors.

2.2 Market

The COVID-19 outbreak had a devastating impact on the passenger aviation demand (Bolić and Ravenhill, 2021, pp.448-451; Liu, Lim and O'Connell, 2021, p.103298). In Fig. 2.2 can be seen that the number of air passengers, globally, plummeted in 2020. Since 2020, more than 60 airlines ceased operations (31 in 2020, 19 in 2021, 12 in 2022, and 3 until February 2023). These bankruptcies were accelerated by the pandemic, but hardly were caused only by it. In 2018, 18 global airlines went burst and, in 2019, another 34. Most of them were small airlines that could not compete with the big players, like Ryanair, that are increasing their market share year after year.

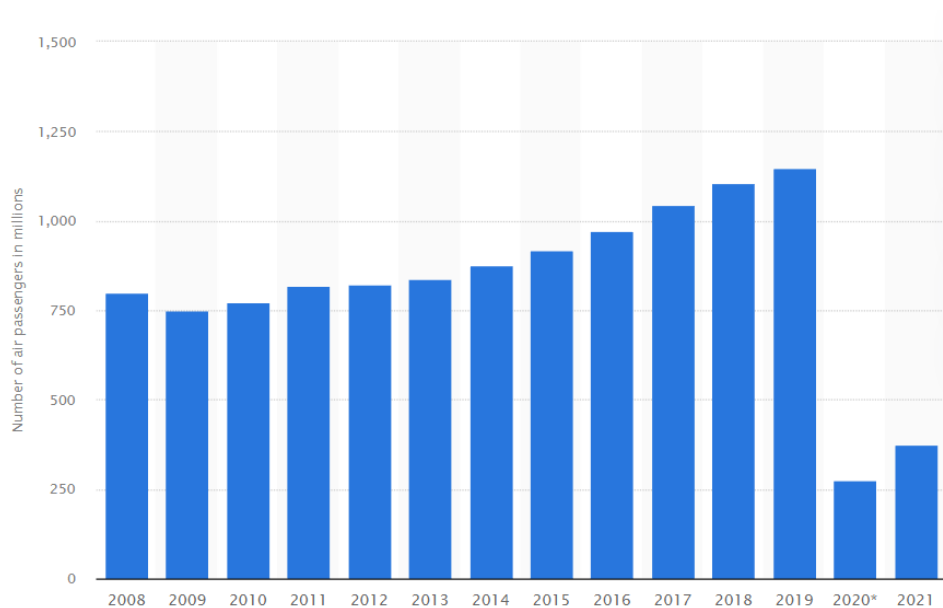


Fig. 2.2 Passengers demand in European market from 2008 to 2021.

Even though companies in Europe are still working to restore their pre-pandemic flight schedules, the market has already stabilized. Teneo, a global CEO advisory firm, published a report in September 2022 forecasting the passenger aviation demand for 2022-2024. The Fig. 2.3 shows this forecast, relative to 2019 demand. The demand in Europe is expected to recover better than in other regions of the globe.

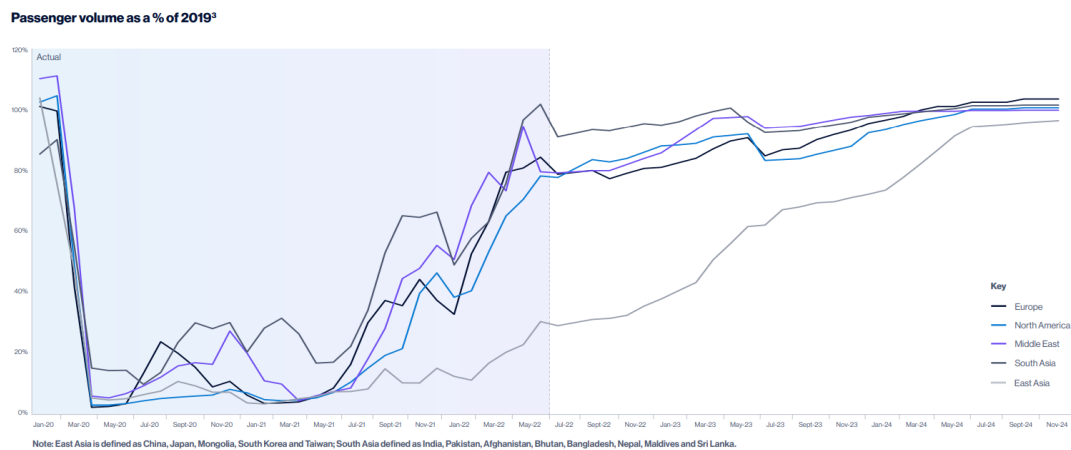


Fig. 2.3 Forecast of passenger aviation demand for 2022-2024.

Some cautions are still advised. Geopolitical factors (e.g., war in Ukraine) impacted in fuel prices, and in customers travel plans (Fig. 2.4). According to European Transport Commission (ETC), only 53% of people did not change travel plans due to the war until 2022.



Fig. 2.4 People who changed travel plans due war in Ukraine.

Other challenges faced by this market are the household disposable income and the number of available employees. European market expects a decrease in customers that are most likely to exit the market due to a reduction in their disposable income. Also, the market already suffers from lack of employees in aviation industry.

2.3 Proposed solution

A BI and BA project on Ryanair passenger satisfaction would involve the analysis of data related to customer experiences, preferences, and feedback throughout the passenger journey. The scope of processes to be considered may include:

- **Data Collection:**

Gathering relevant data from various sources, such as customer surveys, feedback forms, social media platforms, and customer service interactions. This data may include demographic information, flight details, booking patterns, onboard services, and complaints.

- **Data Integration and Cleansing:**

Integrating data from dissimilar sources and ensuring its quality and accuracy. This process involves removing duplicate records, standardizing data formats, and resolving inconsistencies.

- **Data Analysis:**

Applying various analytical techniques to the collected data to identify patterns, trends, and correlations. This analysis may include segmenting customers based on demographics, analysing satisfaction scores, identifying factors impacting satisfaction, and comparing performance across different flight routes or time periods.

- **Customer Journey Mapping:**

Mapping the entire passenger journey, from flight search and booking to pre-flight, in-flight, and post-flight experiences. This helps identify touchpoints where passenger satisfaction can be improved.

- **Key Performance Indicators (KPIs):**

Defining relevant KPIs to measure and monitor passenger satisfaction. Examples may include on-time performance, baggage handling, in-flight amenities, staff behaviour, and complaint resolution time.

- **Predictive Analytics:**

Utilizing predictive modelling techniques to forecast passenger satisfaction levels based on historical data and identify potential areas of improvement. This can help Ryanair proactively address issues and enhance customer experiences.

- **Reporting and Visualization:**

Presenting the analysed data and insights in user-friendly dashboards, reports, and visualizations. This allows stakeholders to easily understand and act upon the findings.

- **CRM Integration:**

By integrating the passenger satisfaction dataset with the CRM system, the airline can better understand the preferences and behaviours of their customers. This can help the airline personalize their offerings and provide a better customer experience.

- **CRM Analytics:**

Using CRM analytics, the airline can analyze customer data to identify trends and patterns. For example, they can analyze customer complaints to identify the most common issues and take steps to address them. They can also analyze customer feedback to identify areas where the airline is doing well and areas where they need to improve.

CHAPTER 3

Management Methodologies

To help organizations make data-driven decisions by gathering, analysing, and presenting relevant information, some methodologies have emerged as effective tools. One useful combination of these tools is SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis plus BSC (Balanced Scorecard). SWOT analysis provides a comprehensive overview of a company's internal and external environment, identifying areas that need improvement, and opportunities for growth (Chang and Huang, 2006, pp.158-169; GURL, 2017), while BSC assists creation of a comprehensive set of performance metrics. BSC provides a balanced view of a company's performance by measuring and tracking four perspectives: Financial, Customer, Internal Processes, and Learning and Growth.

Both methodologies provide a holistic view of the business. Even if a project scope is limited to one field, having this overall view is important to consider the effect of now applied methodologies and technologies in the whole business. By combining SWOT and BSC, we gain a complete picture of the company current situation, identify areas for improvement, and track progress towards its goals.

3.1 SWOT Analysis

The SWOT analysis (Pickton and Wright, 1998, pp.101-109; Helms and Nixon, 2010, pp.215-251) was applied in this project to identify the internal and external factors (Krishnakumar and Baby, 2011, pp.39-50) that impact Ryanair. Fig. 3.1 shows the main points of the analysis and a more detailed explanation of each are given below.

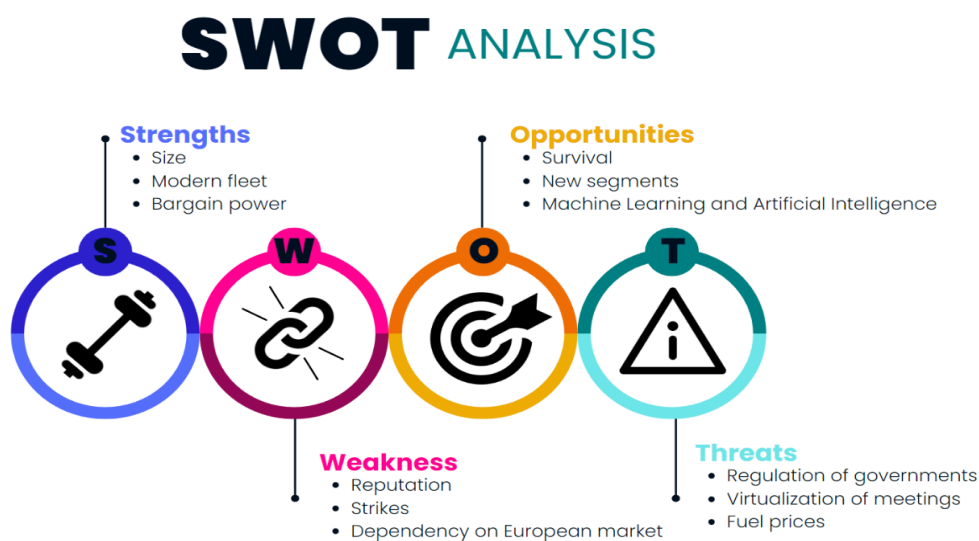


Fig. 3.1 SWOT analysis applied to Ryanair.

3.1.1. Strengths:

- **Size:**

Ryanair is the biggest low-cost airline company in Europe and the airline with most routes in the world.

- **Modern fleet:**

Ryanair operates with modern Boeing models aircrafts. The company uses models that are known by their efficiency, meaning keeping the fuel consumption as low as possible.

- **Bargain power:**

Ryanair can pressure airports to reduce landing fees since the company determine where to fly. Moreover, the business model permits that Ryanair uses fewer demanding spaces in airport's runaways, e.g., farthest boarding and disembarking points.

3.1.2 Weaknesses:

- **Reputation:**

Ryanair scores high when it comes to customer dissatisfaction. Many issues are reported by passengers, like surcharges, uncomfortable seats, and bad customer service. The company's refund policy Is also target of passenger's criticism.

- **Strikes:**

The fierce pursuit of reducing costs also influences the wages provided by Ryanair. This along with airline industry workers' work conditions can result in new strikes, as had seen before (Efthymiou *et al.*, 2021, P.101997).

- **Dependency on the European market:**

Since Ryanair operates flights almost entirely in Europe, any local crisis reflects on company's profits. By May 2023, i.e., many of its flights were cancelled due to air traffic controller (ATC) strikes in France this year.

3.1.3 Opportunities:

- **Survival:**

Ryanair, as many other companies, went through many crises, some that affected specially airlines: the Gulf Wars, the attacks of eleventh September, financial crisis of 2008, and, more recently, the COVID-19 pandemic. Numerous airlines have gone out of business in which of these occasions, but Ryanair is one of the remnants. Every competitor less is an opportunity more.

- **New segments:**

As a low-cost company, the first and medium classes are, at least partially, unexplored by Ryanair. While other premium companies in the sector need to adapt to low-income times, Ryanair can use a windowlike this to expand its business.

- **Machine Learning and Artificial Intelligence:**

The current growth of artificial intelligence and machine learning tools can have a synergistic effect with Ryanair's strategy. These tools are revolutionizing the cost reduction of diverse types of operations (Kaakai et al., 2022, 01-15-02-0009).

3.1.4. Threats:

- **Regulations of governments:**

Ryanair is subject to many state regulations, including, but not restricted to, Ireland's, United Kingdom's, and European Union's.

- **Virtualization of meetings:**

The COVID-19 pandemic catapulted advances in video meetings around the globe. Advances in this area jeopardize revenue coming from business passengers.

- **Fuel prices:**

Fuel expenses make up a large part of Ryanair operational costs. Any rise in the fuel prices can bring troubles. The Russia's invasion of Ukraine caused a rise in jet fuel prices in Europe, for example, and this reflects in the fares.

Two factors of this analysis are directly connected to customers: reputation as a weakness and new segments as an opportunity. Both can be addressed with an improvement of customer management. Ryanair needs to address its weaknesses, particularly concerning customer service, while capitalizing on opportunities for growth and innovation. The company must also adequately navigate external threats, such as economic downturns, health crises, and competitive pressures, to sustain and improve its market position.

3.2 BSC

The BSC is important to keep track of metrics from a company. For Ryanair, this study identified some key performance indicators for each perspective. The Fig. 3.2 summarizes the perspectives from the BSC for Ryanair.

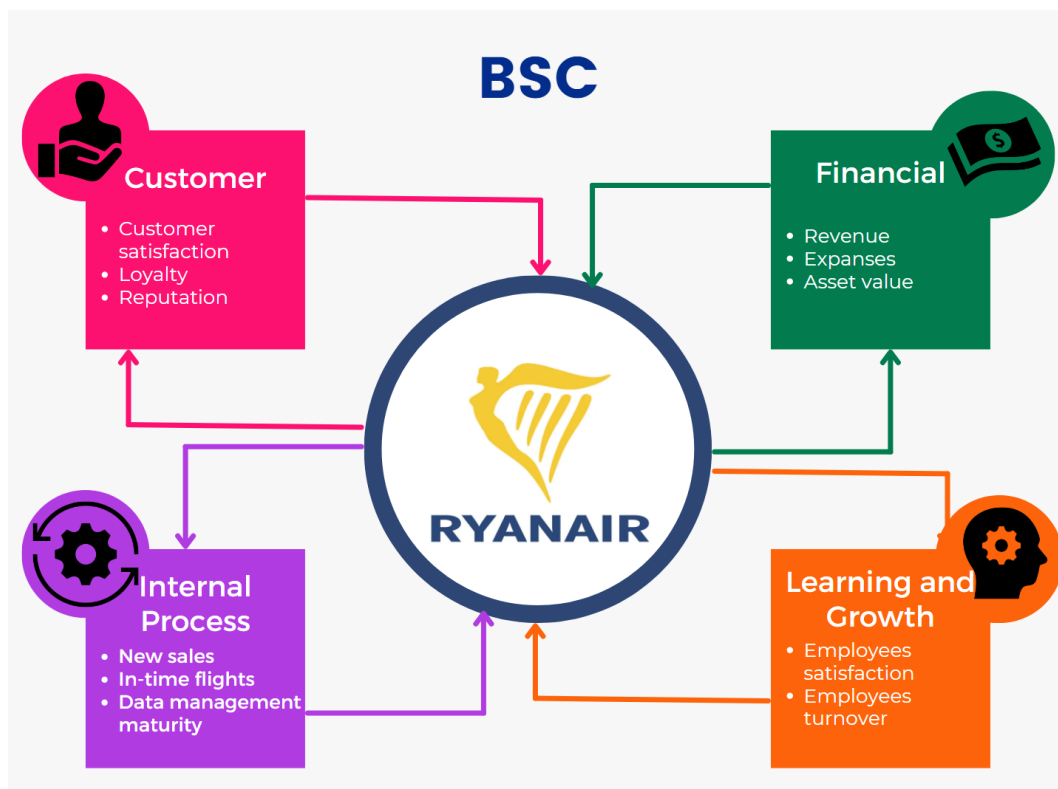


Fig. 3.2 BSC applied to Ryanair.

3.2.1. Customer perspective:

- **Customer satisfaction:**

The main point of implementing a CRM system is to increase the customer satisfaction and consequently increase company's revenue.

- **Loyalty:**

Retain customers is a key process to maintain and growth market share and it is an indirect indicator of customer satisfaction.

- **Reputation:**

Improving the company's reputation might lead to new customers.

3.2.2. Financial perspective:

- **Revenue:**

Increase revenue by implementing the CRM system and enhancing customer satisfaction and retention.

- **Expenses:**

Reduce operational costs by automating processes through the CRM system and optimizing resources.

- **Asset value:**

Improve asset value by investing in technology, infrastructure, and maintenance of the aircrafts.

3.2.3. Internal process:

- **New sales:**

Increase sales by better understanding customer needs and preferences through the CRM system and providing personalized offers.

- **In-time flights:**

Improve on-time performance by using data analysis to optimize flight schedules and resources.

- **Data management maturity:**

enhance data management capabilities by implementing a robust CRM system that enables efficient data collection, analysis, and utilization.

3.2.4. Learning and Growth:

- **Employees satisfactions:**

important aspect to prevent turnovers and guarantee better performance.

- **Employees turnover:**

this should be as low as possible since the availability of employees in the aviation sector is an issue in Europe.

CHAPTER 4

System and Database Design

To streamline our data management process, we utilized the flexibility of CSV files to extract relevant data. We then efficiently loaded this data into PostgreSQL using the powerful PGAdmin tool. This enabled us to establish a robust and secure database environment, ensuring the integrity and accessibility of our information.

To leverage the insights gained from this data, we established a seamless connection between PostgreSQL and Power BI. This integration allowed us to harness the advanced visualization capabilities of Power BI, enabling us to create dynamic and visually appealing dashboards. These dashboards provide a comprehensive overview of our data, empowering us to make informed decisions and gain valuable insights.



Fig. 4.1 System and database design

Recognizing the significance of customer satisfaction, we took proactive steps to enhance our customer service processes. We loaded the same CSV file containing customer feedback into Salesforce Service Cloud, a comprehensive platform for managing customer interactions. Leveraging the capabilities of Salesforce, we utilized workflow triggers to automate email notifications to passengers based on their satisfaction level. This personalized approach demonstrates our commitment to addressing customer concerns and ensuring a positive experience.

By implementing these integrated systems and workflows, we have not only streamlined our data management practices but also strengthened our ability to analyze customer feedback and effectively engage with passengers. This approach enhances our customer satisfaction efforts, ultimately fostering long-term customer loyalty and driving business success.

PART – 2 IMPLEMENTATIONS

CHAPTER 5

Technologies Implemented

There are many players in the market that provides solutions for implementing CRM and data-vis. Some offer both. To make an informed decision, comparisons between the potential providers were carried out using a demo of the SelectHub solution for software selection. This tool provides clear and detailed comparison report with scorecards for each provider based on some characteristics of the contractor company. For this case, Ryanair has more than 10,000 employees and figures in the Travel, Transportation & Logistics Industry.

5.1 CRM solutions

The evaluation of CRM solutions was made prioritizing four modules: Accounts and Contacts Management, Customer Communication Management, Dashboards and Reports, and Sales Performance Management. The products compared were Salesforce Sales Cloud, Pega CRM, Dynamics 365 For Sales, Oracle CX Sales Cloud, and Sugar Sell. In Figure 5.1 is the comparison report highlights. The overall score Salesforce is the highest, followed close by Dynamics 365. Salesforce is the best for this implementation, not just because the overall score, but judging by the four modules prioritized here, Salesforce has better scores than Dynamics in two. To make a final decision, Salesforce scores higher in Customer Communication Management, since it provides good SMS, email, and calls management and schedule. It worth notice that Salesforce scores 100/100 for Training, which is paramount due the high costs of training in big companies.

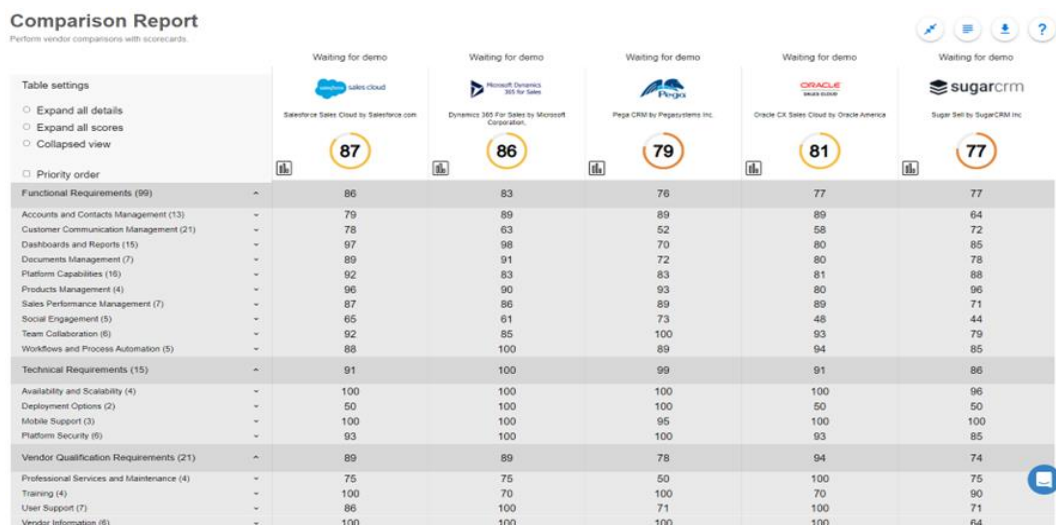


Fig. 5.1 Comparison report between main CRM providers.

Salesforce is a leading provider of cloud-based software solutions that empower businesses to enhance their customer interactions and achieve greater success. With a focus on finding new prospects, increasing sales closures, and delivering exceptional customer service, Salesforce offers a comprehensive suite of products known as Customer 360.

Customer 360 brings together various functionalities, including sales, service, marketing, commerce, and IT, into one integrated platform. By consolidating customer information, it provides a holistic view of each customer, enabling businesses to develop deeper relationships and foster growth. This unified approach benefits both customers and employees, fostering stronger connections and improved collaboration across teams.

Through Salesforce's suite of products, businesses gain access to a range of powerful tools and features. These include advanced analytics, personalized marketing campaigns, streamlined sales processes, intuitive customer service capabilities, and seamless e-commerce experiences. By harnessing these resources, organizations can optimize their operations and deliver exceptional value to their customers.

The cloud-based nature of Salesforce's software ensures flexibility, scalability, and accessibility. Businesses can securely access and manage customer data from anywhere, enabling remote work and facilitating real-time collaboration. This empowers teams to respond swiftly to customer needs, make informed decisions, and deliver personalized experiences at a scale.

In summary, Salesforce is a renowned provider of cloud-based software solutions that enable businesses to elevate their customer interactions. Customer 360, their comprehensive suite of products, brings together various teams and functions, providing a unified view of customer information. With advanced tools and a focus on enhancing relationships, Salesforce empowers organizations to drive growth, improve collaboration, and deliver exceptional customer experiences.

Choosing Salesforce Service Cloud for Ryanair's customer service department offers several compelling reasons:

- **Cost Efficiency:**

Salesforce Service Cloud operates on a cloud-based subscription model, eliminating the need for upfront infrastructure investments and reducing hardware and maintenance costs. Ryanair can leverage the scalability of the cloud, paying for only the resources and user licenses needed, which can lead to cost savings compared to on-premises solutions.

- **Predictable Pricing Structure:**

Salesforce offers transparent pricing plans with clear subscription options. Ryanair can choose the most suitable plan based on its specific requirements and budget. The subscription-based model allows for predictable monthly or annual costs, enabling better financial planning and cost control.

- **Return on Investment (ROI):**

Salesforce Service Cloud's robust feature set and automation capabilities can drive efficiency and productivity gains in Ryanair's customer service department. By streamlining processes, reducing manual tasks, and providing a unified view of customer data, agents can handle more inquiries and provide better service, ultimately leading to improved customer satisfaction and loyalty. The resulting ROI can outweigh the initial investment in the platform.

- **Reduced IT Overhead:**

With Salesforce Service Cloud, Ryanair can rely on Salesforce's infrastructure and technical expertise, reducing the need for dedicated IT staff and resources to maintain and support the system. Salesforce handles software updates, security patches, and system maintenance, allowing Ryanair's IT team to focus on other strategic initiatives.

- **Flexibility to Scale:**

As Ryanair's customer service needs grow, Salesforce Service Cloud can easily scale to accommodate increasing service demands without incurring significant additional costs. Ryanair can add or remove user licenses and expand the usage of the platform as required, providing flexibility and cost control.

- **Cost-saving Integrations:**

Salesforce Service Cloud offers pre-built integrations with a wide range of popular business applications and systems. This allows Ryanair to leverage existing investments in tools such as telephony systems, email providers, or survey platforms, eliminating the need for costly custom integrations or duplicate subscriptions.

By considering these cost-related points, Ryanair can benefit from the cost efficiency, predictable pricing structure, ROI potential, reduced IT overhead, scalability, and cost-saving integrations offered by

Salesforce Service Cloud, making it an attractive choice for optimizing its customer service operations while keeping costs in check.

5.2 Data visualization solutions

For Data-vis implementation, five products were analysed: Tableau, Power BI, Oracle Analytics Cloud, Dundas BI, and SAP Analytics Cloud. The most relevant modules to compare were Dashboarding and Data Visualization and Data Querying. The overall comparison report can be seen in Figure 5.2. It can be noticed that Power BI presents consistent scores high in all macro requirements, so it was chosen as the best choice for this implementation.

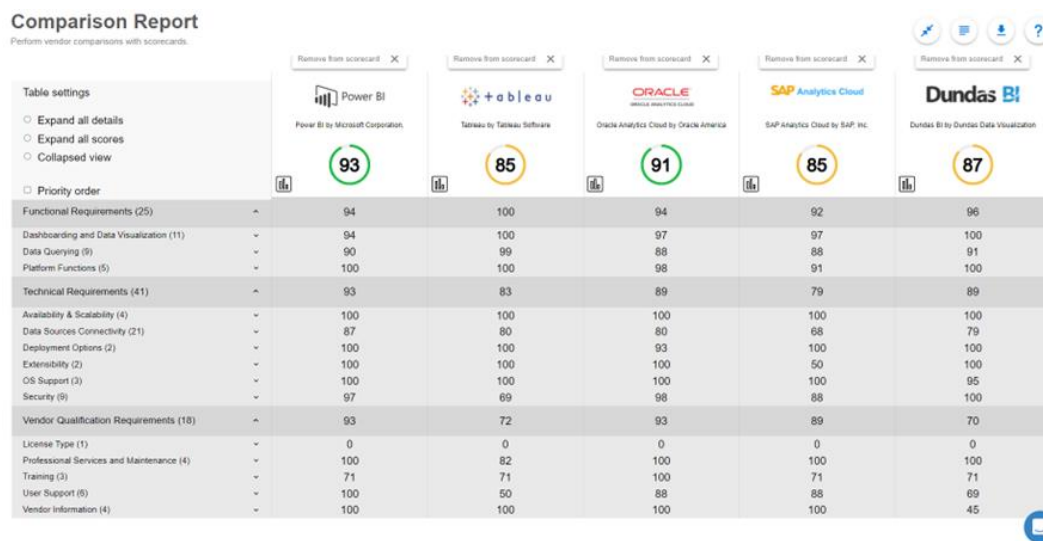


Fig. 5.2 Comparison report between main data visualization providers.

Power BI is a very popular Microsoft solution (Ferrari and Russo, 2016) for Dashboarding and Data Visualization. It is behind the competitors only regarding Animations, because it does not provide fully out of the box to customize the dashboard in this sense. When it comes to Data Sources Connectivity, it scores higher than all the others, since presents great connectivity with popular accounting platforms like Intuit Quickbooks, FreshBooks, Sage, etc. Moreover, and with high priority to this project, the solution connects to popular CRM platforms like SalesForce, HubSpot, Insightly, FreshSales, etc. with high score. To conclude, Power BI is the only solution, among the five analyzed, that scores 100/100 on User Support.

Hereafter, there is a list of benefits of using Power BI, provided by SelectHub:

- **Live 360-Degree View of Data:**

Stay in the loop about crucial business insights. Power BI's dashboards update in real time from all data sources.

- **Visualize Data Trends:**

Create intuitive visualizations to present data after it is parsed, structured, and analyzed. This can be in the form of graphs, charts, maps, or other graphic depictions that make interpreting data quicker, easier, and more impactful.

- **Make Data-Driven Decisions:**

Put the data visualizations to work — analyze not only what happened, but also why, how, and where. Use data to improve business processes and customer interactions, adjust business plans and make forecasts.

- **Ease of Use:**

Encourages user adoption towards a data driven culture. Power BI simplifies self-service BI with built-in features similar to Excel and other Microsoft technologies — feel right at home with functionalities like drag-and-drop design, resizing and copy-and-paste.

- **Collaborate Across Organization:**

Share reports and collaborate on decisions quickly; save time otherwise spent on inefficient meetings or jumping from one platform to another. Integrates seamlessly with many commonly used business tools to promote teamwork throughout departments.

- **On-the-Go Access:**

Access visual analytics on the go with Power BI. Stay connected to and share insights from any device, with mobile-optimized reports and a mobile app.

- **Secure Insights:**

As one of the largest, most well-known names in the software industry, Microsoft backs up their reputation with security and protection for businesses' sensitive information in Power BI. Visit the

Trust Center to learn more about exactly how all of Microsoft's solutions — including Power BI — meet industry security standards and ensure regulatory compliance. Keeps information secure from the original data source all the way up through the report sharing and collaboration features.

- **Free Version:**

Power BI goes beyond a free trial with a perpetually free Desktop version, suitable for creating reports and data visualizations on an individual basis.

- **Scalability and Affordability:**

Power BI can scale from the individual to the enterprise level, while only paying for exactly what is necessary — choose the right option from versions varying in capabilities and pricing.

CHAPTER 6

Customer Relationship Management

Customer Relationship Management is a strategic approach utilized by businesses to effectively manage their interactions and relationships with customers (Chen and Popovich, 2003, pp.672-688; Payne, 2006). It encompasses methodologies, software, and internet capabilities that enable organizations to organize customer-related information in a structured manner. By doing so, businesses can enhance their understanding of customers, improve customer satisfaction, and drive long-term profitability (Rigby and Ledingham, 2004, pp.118-130).

CRM systems provide a centralized database that stores comprehensive customer data, including contact information, purchase history, preferences, and interactions. This database facilitates easy access to relevant customer information by various stakeholders, such as sales teams, customer service representatives, and management personnel (Nguyen, Sherif and Newby, 2007, pp.102-115). It enables them to make informed decisions, personalize interactions, and deliver targeted marketing campaigns.

The integration of sales, customer service, marketing, and other customer-facing functions is a key aspect of CRM. By aligning these departments and fostering collaboration, businesses can create a seamless customer experience throughout the entire customer journey. This holistic approach ensures consistent messaging, efficient issue resolution, and timely response to customer inquiries, resulting in increased customer satisfaction and loyalty.

Moreover, CRM leverages technology and the internet to maximize customer relationships. Through customer portals and self-service options, organizations empower customers to access relevant information, track orders, and initiate support requests, reducing reliance on traditional communication channels. CRM systems also enable automation of routine tasks, streamlining processes and freeing up resources to focus on high-value customer interactions.

One of the primary objectives of CRM is to optimize customer value while minimizing costs. By leveraging customer data and analytics, businesses can identify their most profitable customers and tailor their offerings accordingly. This includes developing targeted marketing campaigns, cross-selling or upselling products and services, and providing personalized customer experiences. Simultaneously, CRM helps reduce costs by improving operational efficiencies, enhancing customer retention, and reducing customer churn.

In summary, CRM is a strategic approach that combines methodologies, software, and internet capabilities to effectively manage customer relationships. It integrates departments, leverages technology, and maximizes customer value while reducing costs. By adopting CRM principles, businesses can build strong customer relationships, drive customer satisfaction and loyalty, and ultimately achieve sustainable growth and profitability (Kostojohn, Paulen and Johnson, 2011).

6.1 Implementation Model

To effectively manage and analyze customer feedback, we have implemented a customized object called "Customer Satisfaction" within the Salesforce platform. This object serves as a repository for storing all the feedback provided by passengers. The data captured includes various factors such as gender, customer type, age, type of travel, class, flight distance, inflight Wi-Fi service, departure/arrival time convenience, ease of online booking, gate location, food and drink quality, online boarding experience, seat comfort, inflight entertainment, on-board service, legroom service, baggage handling, check-in service, inflight service, cleanliness, departure delay in minutes, arrival delay in minutes, and overall satisfaction rating.

By utilizing this data, we can generate comprehensive reports and dashboards that provide valuable insights into customer satisfaction levels. These reports and dashboards enable management to identify areas of improvement and weaknesses from the customer's perspective.

Upon the creation of a feedback record, an automated process is triggered. Passengers receive a personalized thank you email as an acknowledgement of their feedback. In cases where a customer expresses dissatisfaction, a 20% discount offer is sent to apologize for their experience. Additionally, a customer service case is automatically created to address the concerns raised and ensure a better experience for the customer in the future.

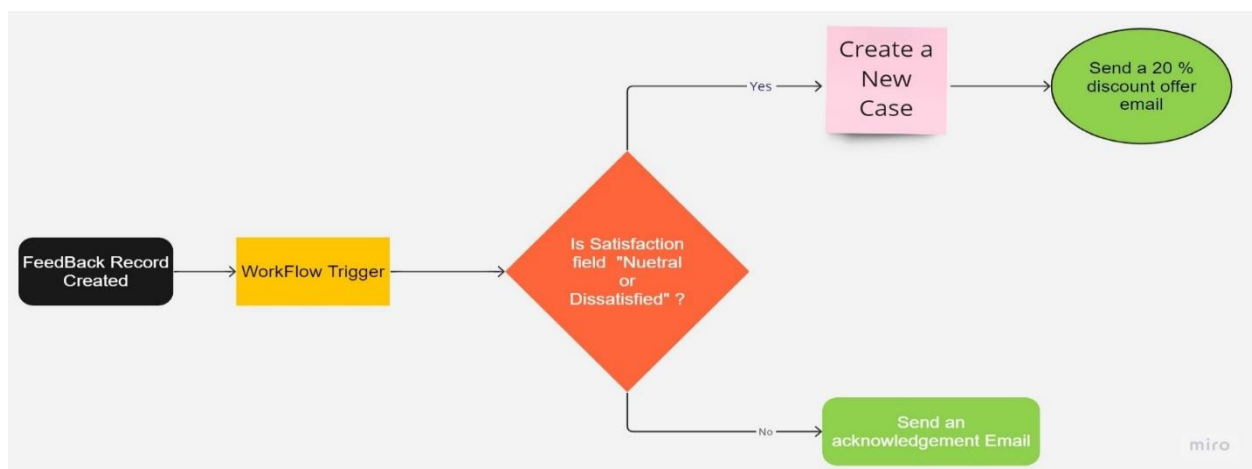


Fig. 6.1 Implementation Model.

The feedback records, along with the corresponding reports and dashboards, serve as a valuable resource for management to gain a deeper understanding of areas requiring improvement in terms of customer satisfaction. By leveraging this information, Ryanair can identify trends, patterns, and common issues, allowing them to take proactive measures to enhance the overall customer experience.

To maintain a consistent brand image, an email template has been created using Ryanair letterhead. This ensures that all communication sent to customers aligns with the company's branding guidelines and presents a cohesive and professional appearance.

In summary, the custom object "Customer Satisfaction" stores passenger feedback data, which is utilized to generate reports and dashboards. The automated processes of sending thank you emails, offering discounts for dissatisfied customers, and creating customer service cases contribute to a proactive and customer-centric approach. The insights derived from this data aid management in identifying areas for improvement and strengthening the overall customer experience.

CHAPTER 7

PowerBi Dashboard

In the modern era of data-driven decision-making, airlines have recognized the immense value of leveraging data to improve operational efficiency, enhance customer experiences, and drive strategic growth (Cooper et al., 2016 pp. 1-23; Allio, 2012, pp.24-31). To accomplish these objectives, we have used database management system, Postgres (PostgreSQL) for efficient data loading, storage, and retrieval. It is an open-source, object-relational database management system renowned for its robustness, flexibility, and scalability. Data collected from various sources have been loaded into postgres as tables in BIBA_Airline database.

The majority of data was collected from Statista which had Ryanair's original operations and expenses insight. Airline customer satisfaction data was created using feedback given by passengers on various travel sites. Ryanair operational routes data is official data which contains values of latitudes, longitudes, origin country, destination country, ICAO country codes, etc. Additionally, the data for new destinations which are planned for operation this summer has been collected from official website along with base fare. The data gathered has been loaded into excel sheets and cleaned before loading into postgres (Fig. 7.1).

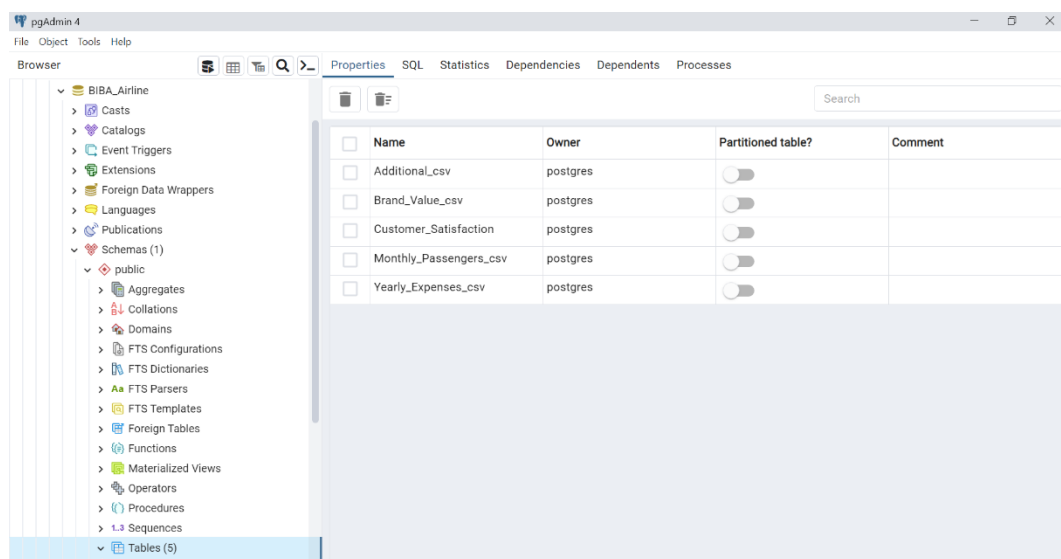


Fig. 7.1 Dataset in postgres.

Power BI, a leading business intelligence and data visualization tool is used in integration with Postgres. Data is extracted, transformed, and visualized in a user-friendly and interactive manner. The integration between Postgres and Power BI is achieved through direct database connections. Four dashboards have been created for achieving insights to maintain customer loyalty in Ryanair.

7.1 Europe’s Leading Airlines

The purpose of the dashboards is to provide comprehensive insights into the European competitors of Ryanair within the aviation industry. By analyzing various metrics and indicators, these dashboards offer a valuable understanding of Ryanair's performance relative to other airlines in the region. As we can see in Fig. 7.2, in 2021, Ryanair recorded the highest number of passengers carried among all the airlines considered in the comparison. This significant achievement signifies Ryanair's success in attracting a substantial customer base and highlights its position as a leading player in the European market. However, it is crucial to note that the brand value of Ryanair, as indicated by consecutive yearly data, does not surpass that of another airline.

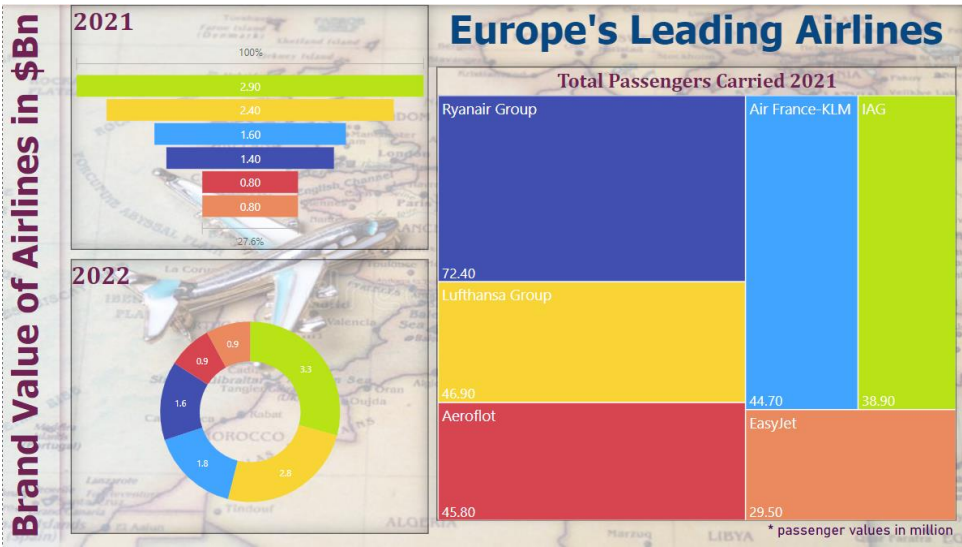


Fig. 7.2 View comparing Ryanair and its competitors.

Despite Ryanair's remarkable passenger numbers, the brand value of the company, when evaluated over two years, reveals that IAG holds a superior position in terms of brand recognition and perception. This insight suggests that although Ryanair excels in attracting passengers, the competitor airline maintains a stronger and more valuable brand image. The dashboards present all numerical values in million euros, ensuring consistency in the unit of measurement for comparison across European airlines. In terms of passenger volume, the Ryanair Group showcases a staggering 72.40 million passengers choosing their airline to travel to their respective destinations. This figure surpasses even the passenger carried count of IAG, which is recognized as the top airline in terms of brand value for both the assessed years.

7.2 Passenger and crew insights

The dashboard presented in Fig. 7.3 provides an overview of Ryanair's performance in terms of passenger volume and employment from the year 2011 to 2022. All the values displayed in this dashboard are in millions, reflecting the prominent position held by Ryanair in the European aviation industry. The airline boasts one of the largest fleets and employee bases, solidifying its stature in the market. The top employment graph highlights the number of individuals employed by Ryanair, showcasing the company's commitment to generating job opportunities. Notably, even during the challenging period of the COVID-19 pandemic, when numerous airlines were forced to halt operations and implement layoffs, Ryanair exhibited resilience by continuing to recruit cabin crew and pilots. This recruitment activity is reflected in the 2021 employee share graph, further validating the airline's dedication to sustaining employment levels during a time of crisis.

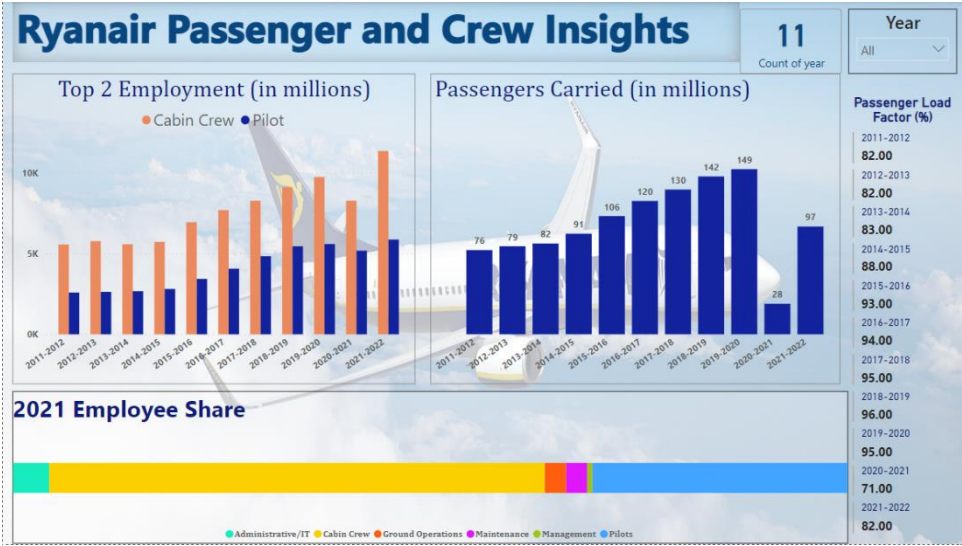


Fig. 7.3 View of passengers and crew from Ryanair.

In terms of passenger numbers, the dashboard reveals a significant decline in 2020-2021 due to the global pandemic. The unprecedented challenges posed by the COVID-19 outbreak resulted in a substantial decrease in the number of passengers carried by Ryanair. However, the subsequent year shows a promising recovery and positive growth for the airline, indicating its resilience and potential for surpassing pre-pandemic levels. This recovery signifies a scope for improvement and a positive trajectory towards returning to the passenger volumes experienced before the COVID-19 era.

Additionally, the dashboard provides insights into the passenger load factor, which measures the percentage of available seating capacity utilized by passengers. Notably, in the years 2018-2019, Ryanair achieved an all-time high passenger load factor of 96%, indicating efficient capacity utilization.

However, the onset of the pandemic resulted in a significant drop in the load factor, plummeting to approximately 71%. This decline can be attributed to travel restrictions and reduced demand during the COVID-19 crisis.

In conclusion, the presented dashboard showcases Ryanair's performance in terms of passenger volume and employment over the assessed period. Despite facing challenges posed by the global pandemic, the airline exhibits resilience, evidenced by its recruitment efforts during a time when other airlines were forced to implement layoffs. While the passenger carried count experienced a temporary decline, the subsequent year displays positive growth, indicating a path towards recovery and exceeding pre-pandemic levels. Moreover, the passenger load factor experienced fluctuations due to the impact of COVID-19, but Ryanair's historical achievement of a high load factor demonstrates its capacity utilization efficiency.

7.3 Passenger satisfaction analysis

In Fig. 7.4 we can see that more than 57% of the customers are not satisfied with the services that they were provided with, gender-based distribution gives us a brief idea on how the matrices affect different genders, Looking at the type of travel business travel has more count for satisfaction, this may be due to most of the business travels are paid the businesses so the customer does not expect much, but on the other hand in personal travel the customer is spending their own money so they expect the services better and expect a bang for their buck due to which we can see the count of neutral or dissatisfied more in personal travel. Now comes the main talking point of this dashboard the distribution of customer-based satisfaction that is satisfaction in loyal and disloyal customers, we can see that the numbers of neutral or dissatisfied is higher in loyal customers, this may be a huge blow to the company as these are the customers who always choose the airline, if the customers who prefer your airline are not satisfied may start to lean with different airlines that may provide better services, retaining loyal customers should be the main part of the airline rather than converting disloyal customers to loyal, because if the loyal customer are happy/satisfied this condition will directly lead to disloyal customers becoming loyal customers.



Fig. 7.4 View of Ryanair’s customers feedbacks.

We can see the total rating given by the users. These are some of the important features that are more important compared to others, the dashboard also contains the average of arrival and departure delay in minutes.

As we can see that the average delay in departure and arrival is less, as this is one of the strongest filed for the airline where they have already done the things and taken the necessary precautions to avoid delay as delay is one of the top factor that decides the sentiment of the customer, if there is any delay there is a very Seaver impact on customer sentiment and satisfaction.

The rating gives us an insight on the services provided by the company, the highest rated is Inflight services, which has a significant impact on customer sentiment as better service is a must.

Flight reservations may be made easily, and the vacation planning process is streamlined using airline online booking. Customers may check flight schedules, evaluate pricing, and reserve their desired tickets at their convenience using this digital platform. Having a bad rating in this section not only hurts the customer sentiment but also hurts the airline as people are having a hard time booking their tickets online. Having a better supported website will help the company as customers will have easy time booking tickets.

7.4 New destination routes

In Fig. 7.5 we can see the top 17 departure cities, London being the top, as we can see Ryanair is a top player in Europe where it covers most or all the Europe.



Fig. 7.5 View of Ryanair routes.

The above section of the dashboard demonstrates the new destinations added by the company, most of the destinations are in Europe, the company already has a strong connection in Europe by adding the above destinations they are trying to strengthen their grip on Europe as the company has the vision of being the only major airline company in Europe, as their core market is Europe. The Dashboard also demonstrates the fare prices for the new destinations (For one way)

From this dashboard we can see that after suffering loss during Covid lockdown, the company is not scared to make moves, and increase connectivity in Europe.

7.5 Passenger and operations insights

The presented dashboard offers valuable insights into the monthly passenger volume and operational aspects of Ryanair spanning the years 2011 to 2022. It provides a comprehensive overview of the airline's performance and sheds light on key trends and fluctuations in passenger numbers. One crucial aspect highlighted in the dashboard is the total operating expenses incurred by Ryanair. In 2019-2020, the

airline experienced a significant increase in operating expenses, reaching a new high. However, the following year witnessed a sharp decline, approximately five times lower, primarily attributed to the adverse impact of the COVID-19 pandemic on the aviation industry. This unprecedented event had a massive disruptive effect, leading to a substantial reduction in operating expenses for Ryanair and other airlines worldwide.

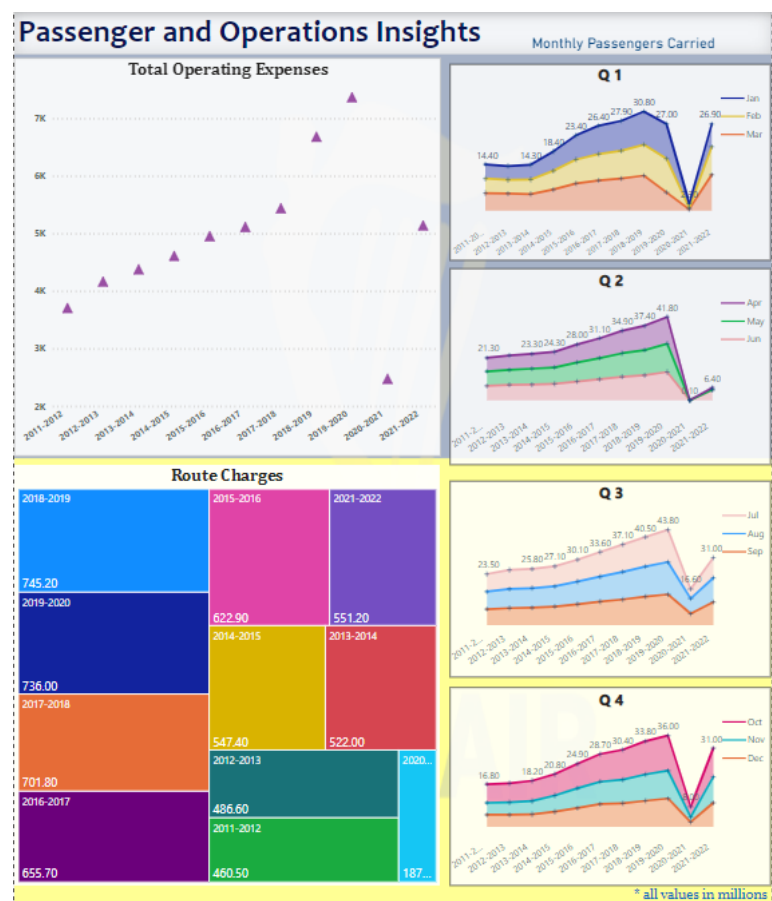


Fig. 7.5 Passenger and operations Insights

Route charges, which airlines are required to pay when operating across various destinations, are also depicted in the dashboard. In 2020-2021, Ryanair encountered the lowest route charge, amounting to approximately 187 million euros. This decline can be attributed to the pandemic's widespread impact, which severely limited or even halted flight operations across multiple locations, resulting in reduced route charges for the airline and its counterparts.

The data on monthly passengers carried over the 11-year period is divided into four quarters to provide a more detailed understanding of passenger volume fluctuations. The first quarter Q1, encompassing January, February, and March, experienced a dip in passengers during the 2020-2021 period due to the

pandemic. However, the following year showed a notable recovery in passenger numbers during the same quarter. Similarly, the second quarter Q2, consisting of April, May, and June, saw a significant decline in passenger volume in 2020-2021. Unlike the first quarter, there was not much recovery observed during this period in the subsequent year, indicating a prolonged impact of the pandemic on travel demand. The third quarter Q3, comprising July, August, and September, experienced a comparatively lower dip in passenger numbers during 2020-2021. This quarter also witnessed a steady recovery in passenger volume in the following year, suggesting a gradual return to pre-pandemic levels. Lastly, the fourth quarter Q4 displayed a substantial decline in passenger numbers in 2020-2021. However, within this quarter, October exhibited the most significant bounce-back compared to November and December, indicating a partial recovery toward the end of the year.

7.6 Annual insights

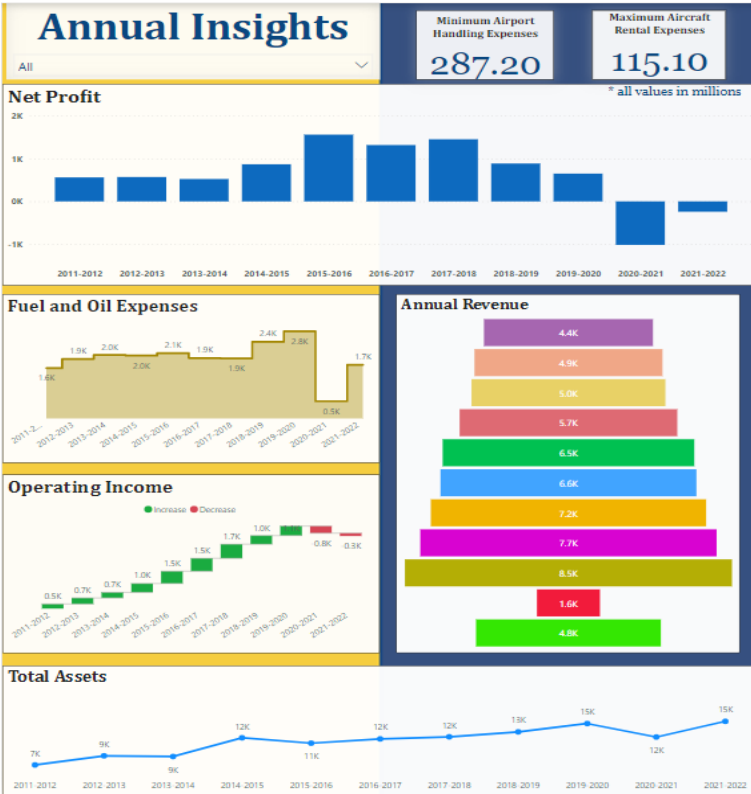
The provided dashboard offers annual insights into the income and expenditure of Ryanair. It presents a comprehensive view of the financial performance of the airline, allowing users to analyze individual years or view multiple years in combination through the slicer function. One crucial aspect depicted in the dashboard is the minimum airport handling expenses incurred by Ryanair for each year. The values displayed in the dashboard are dynamically selected based on the chosen year(s) to provide a comprehensive overview of the expenditure incurred by the airline in airport operations. Similarly, the dashboard also illustrates the aircraft rental expenses, which vary based on the selected year(s). This variable nature of expenses emphasizes the influence of different factors, such as lease agreements and fleet size, on the financial position of Ryanair.

The net profit graph serves as a crucial determining factor when identifying a drop in profit during the 2020-2021 period. This graph showcases the financial performance of Ryanair, highlighting the impact of the COVID-19 pandemic on the airline's profitability. The significant decrease in net profit during this period indicates the adverse effects of the global health crisis on the aviation industry as a whole.

Furthermore, the dashboard displays the fuel and oil expenses incurred by Ryanair. Notably, these expenses experienced a substantial drop during the COVID-19 year due to reduced flight operations and decreased fuel consumption. However, the recovery in fuel prices worldwide, coupled with ongoing geopolitical tensions affecting fuel prices, has resulted in a gradual increase in these expenses in recent years.

The operating income graph reveals a significant decline in operating income, with a continued downward trend in the last two years. This trend underscores the challenging operating environment

faced by Ryanair and the broader aviation industry, potentially influenced by factors such as reduced demand, increased competition, and higher operating costs. Although the annual revenue also experienced a drop, it displayed a recovery to favorable numbers in the 2021-2022 period, indicating a positive trajectory for the airline. This recovery signifies the resilience and potential for improvement in revenue generation, suggesting an opportunity for further growth.



Additionally, the dashboard provides insights into the total assets of Ryanair. While experiencing a decline in the 2020-2021 period due to the impact of the pandemic, the airline's total assets recovered well in the subsequent year, returning to pre-COVID levels. This recovery highlights the ability of Ryanair to adapt and rebound from the challenges posed by the global health crisis.

CHAPTER 8

Project Management

Reflecting on the teamwork exhibited throughout the project, it is evident that the team collaborated effectively and demonstrated a high level of cohesion in working towards achieving our objectives. From the outset, there was a shared understanding of the project goals and a collective commitment to success.

Communication played a pivotal role in our team's ability to work cohesively. Regular meetings were held to discuss progress, address challenges, and share insights. Each team member actively contributed their ideas and perspectives, fostering a collaborative environment where everyone's input was valued. This open exchange of ideas allowed us to capitalize on the diverse skill sets and knowledge within the team, leading to innovative solutions and a well-rounded project outcome.

Furthermore, our team exhibited strong coordination and organization throughout the project. Roles and responsibilities were clearly defined, ensuring that tasks were assigned appropriately, and deadlines were met. Additionally, effective project management tools were employed to track progress and allocate resources efficiently. This level of organization contributed to a smooth workflow and minimized unnecessary delays or conflicts.

Another strength of our team was adaptability. As the project progressed, unforeseen challenges and changes arose. However, the team demonstrated a proactive and flexible approach, adjusting plans and strategies as needed. This adaptability allowed us to navigate obstacles effectively and maintain momentum towards our objectives.

It is important to acknowledge the specific contributions of team members towards achieving the project objectives.

Sarvochch played a pivotal role in the development of the customer satisfaction dashboard, leveraging his expertise in data analysis and visualization. His dedication to creating a user-friendly and insightful dashboard allowed us to effectively track and analyze customer feedback, contributing to our understanding of passenger preferences and overall satisfaction levels.

Rayair flight dashboard, another significant deliverable, was skillfully developed by Sarvoch, providing comprehensive insights into flight-related metrics and performance indicators. His attention to detail and

technical proficiency ensured that the dashboard accurately represented key operational aspects, enabling us to monitor and optimize our airline services.

Nivriti's contribution to creating the Yearly insights and market competition dashboard using Power BI proved invaluable. Her ability to transform complex data into meaningful visualizations and actionable insights empowered our team and stakeholders to make informed decisions and identify trends for strategic planning.

Tulio's role in the project was critical in terms of business understanding and overall project implementation. His focus on management methodologies provided a deep understanding of the industry and business requirements to the team, helped align our objectives and deliverables with the needs of the organization and customers. Tulio's insights and guidance ensured that our efforts remained focused and aligned with the overarching goals.

Deepak's expertise in Salesforce implementation was instrumental in integrating the customer satisfaction data into Salesforce Service Cloud. His meticulous attention to detail and thorough understanding of the platform ensured a seamless implementation, enabling us to effectively leverage Salesforce for automating email triggers based on customer satisfaction levels.

Each team member's specialized skills and contributions were essential in bringing the project to fruition.

In conclusion, the team's strong communication, coordination, adaptability, and supportive environment were instrumental in successfully achieving the objectives of the project.

CHAPTER 9

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